MUHAMMAD JUNAID

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PROFESSIONAL SUMMARY

Full-Stack MERN Developer with hands-on experience building end-to-end web platforms using React, Node.js, Express.js, and MongoDB. Experienced in developing interactive 3D experiences with Three.js and WebGL, integrating LLM APIs, and deploying applications with Vercel and Render. Focused on writing maintainable, secure code and delivering responsive digital experiences.

TECHNICAL SKILLS

Frontend Development:	React, JavaScript (ES6+), Three.js, Tailwind CSS, HTML5, CSS, Redux Toolkit
Backend & APIs:	Node.js, Express.js, REST APIs, JWT authentication, role-based access control, Postman
Databases & Data Modeling:	MongoDB, MongoDB Atlas, Mongoose, schema validation
Version Control & Deployment:	Git, GitHub, Vercel, Render
AI & Integrations:	Groq API, Gemini API, Python, voice-based applications

SELECTED SOFTWARE ENGINEERING PROJECTS

NotesNest — Full-Stack Notes Management Platform

React, Node.js, Express.js, MongoDB, Mongoose Tailwind CSS, Render, Vercel

- Architected a full-stack note management platform using React, Express.js, and MongoDB, implementing 14+ REST API endpoints for note and task CRUD operations.
- Implemented dual-token authentication with short-lived JWT access tokens and HTTP-only refresh tokens, using bcrypt password hashing and automated email verification workflows.
- Built rich-text notes and to-do lists with search, pagination, pinning, favorites, trash and restore workflows, and scheduled reminders.

Interactive 3D Chemistry Laboratory Simulator (Undergraduate Capstone)

Three.js, React, Redux Toolkit, Node.js, Express.js, MongoDB

- Developed a browser-based 3D laboratory simulator with Three.js and React, optimizing mesh geometry and animation loops to maintain approximately 60 FPS during interactive scenes.
- Built a client-side reaction engine that models dynamic titration curves with under 50 ms computation time while synchronizing SVG and Canvas visualizations.
- Designed MongoDB schemas and backend endpoints to store experimental parameters, enabling multi-angle session replay through a student dashboard.

Autonomous Quadruped AI Companion & Control Bridge

Python, PyWebView, Groq API, ESP32-S2 Mini, C++

Architected an asynchronous Python companion service that connects desktop clients to the Groq API, parses voice commands, and triggers robot gait actuation.

Built a PyWebView desktop interface and web bridge for robot control, combining live status monitoring, AI conversation, and motion execution in one interface.

CERTIFICATIONS & HONORS

- **Google AI Professional Certificate:** Applied AI workflows, prompt design, and intelligent systems (Coursera).
- **Google AI Essentials:** Machine learning fundamentals and generative AI models (Coursera).
- **Punjab Educational Endowment Fund (PEEF) Scholarship:** Merit-based academic scholarship awarded for top ranking.

EDUCATION

Government College University Faisalabad (GCUF) — **Expected Aug 2027**

Bachelor of Science in Computer Science (BS CS) — CGPA: 3.80 / 4.00 — **Punjab, Pakistan**

Relevant Coursework: Database Systems, Distributed Computing, Computer Networks, Software Engineering, Data Structures & Algorithms.